

Syed Sabbih Haider Shah

Senior AI Research Engineer

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Summary

Senior AI research engineer with 7+ years of experience designing, training, evaluating, and deploying machine learning systems across scientific forecasting, generative AI, AI security, speech, and computer vision. Experience spans distributed training on NERSC Perlmutter and ORNL Frontier, production inference, geospatial data pipelines, and evaluation designed around real operational failure modes. Combines research ownership with hands-on Python, PyTorch, C++, CUDA, MLOps, and technical leadership.

Core Skills

Machine learning: PyTorch, Transformers, U-Nets, diffusion models, cross-encoders, spatiotemporal forecasting, computer vision, ASR/TTS, model calibration and evaluation

Systems and data: Python, C++, CUDA, NumPy, pandas, xarray, Dask, Zarr, ONNX, NVIDIA DeepStream, Docker, Git, Linux, Slurm

Infrastructure: Distributed multi-GPU training and inference, NERSC Perlmutter, ORNL Frontier, Azure, GCP, streaming and batch pipelines

Professional Experience

Vayuh.ai

Senior AI Research Engineer — Full-time

Jan 2022–Present
United States / Remote

- Train and evaluate spatiotemporal weather models across sub-seasonal, seasonal, and decadal horizons, including billion-parameter research workloads on Perlmutter and Frontier.
- Built peril-specific U-Net workflows over a 1980–2025 atmospheric archive, with diagnostics for spatial agreement, hotspot capture, ranking, and probability calibration.
- Built and audited a 50,000-year, 3 km thunderstorm-wind catalog aligned with hail and tornado simulation years: 155.1 million parent storms, 28.36 billion event-cell rows, and 515 GB of Parquet output.
- Developed catastrophe-model calibration and loss-cost validation using NOAA event and gust records, correlated multi-peril sampling, and FEMA Hazus vulnerability assumptions.

Editable AI

AI Research Lead — Full-time

Feb 2025–Present
Remote

- Demonstrated Hyper-CONUS v2 achieved lower four-lead mean RMSE than HRRR for 2024 2 m temperature and dew point; at observed ± 5 K anomalies, temperature RMSE was 11.2% lower for warm cases and 16.3% lower for cold cases.
- Built the reproducible verification pipeline across 702 matched initialization times, eight forecasting systems, native and common grids, and URMA/MRMS ground truth.
- Built a deterministic severe-weather reporting pipeline that combines MRMS, SWDI, Local Storm Reports, Storm Events, ASOS observations, radar coverage, and explicit evidence scoring into reviewable case packets.

Dashverse

Lead Research Engineer — Full-time

Apr 2023–Feb 2026
United States / Remote

- Designed, implemented, and deployed the AI inference stack for multimodal content-generation products.
- Led data collection, model development, and integration across language, vision, image-generation, and multimodal systems.
- Trained and deployed multiple controllable Stable Diffusion models and led research on next-generation image and text generation.

WeVoz

Lead Machine Learning Engineer — Contract

Jul 2021–Sep 2023
Italy / Remote

- Trained and deployed a custom Italian speech-recognition model achieving 4.1% word error rate and a punctuation model achieving 92% accuracy.
- Optimized inference for low-compute environments and implemented scalable file-based and real-time streaming audio pipelines.
- Generated synthetic noisy speech data to improve robustness under deployment conditions.

Pagarba Solutions

Team Lead, AI, MLOps & Optimization — Full-time

Sep 2021–Jan 2023
United States / Remote

- Led cross-functional teams and end-to-end ML delivery across generative AI, diffusion, computer vision, and mobile inference projects.
- Built and deployed a 30,000-class LEGO recognition system with 96% top-3 accuracy and optimized models for constrained devices.

Additional Experience

STech.ai

Machine Learning Engineer — Full-time

Sep 2019–Jan 2022
Islamabad, Pakistan

- Led an RGB-only facial anti-spoofing system from research through deployment, achieving 97.6% accuracy without depth input.
- Built and optimized real-time NVIDIA DeepStream pipelines, custom plugins, and CUDA-based GPU components.

Teachinguide

Data Scientist — Contract

Jan 2021–Nov 2021
Switzerland / Remote

- Reduced model inference and deployment spend by 70% while implementing automated deployment, data logging, and metric monitoring.
- Deployed revenue forecasting at 4% mean absolute error; course classification at 96% precision and 95% recall; and a Transformer multi-topic classifier at 94% accuracy.

Selected Research and Engineering Projects

Weather forecasting and severe-weather reporting

Scientific ML / HPC

- Built matched forecast comparisons across regional numerical guidance and global AI systems, plus anomaly-conditioned evaluation for warm and cold extremes; published methods and results at editable.ai/forecast.
- Published a property-level hail review using five weather sources, time-aligned radar and report evidence, and a four-page audit packet: editable.ai/evidence.

AI-agent prompt-injection detector

AI security / NLP

- Built a context-aware cross-encoder pipeline with strict separation between 19 training sources and 10 held-out benchmarks; created and audited 13,421 adversarial and hard-negative training pairs across agent, RAG, tool, memory, and multilingual attack surfaces.

Governed multimodal video production runtime

Generative AI / Systems

- Designed a fail-closed pipeline for video, dialogue, Foley, lip sync, compositing, and final assembly with hash-bound evidence, resident multi-GPU batching, targeted retries, 107 Python modules, and 185 automated tests.

Severe-weather event-set generation

Open source / Geospatial ML

- Implemented ensemble hail-magnitude sampling and multi-year Zarr processing workflows for reproducible stochastic event catalogs: github.com/sabbih-shah/eventset-generation.

Education

Master of Engineering, Machine Learning and Image Processing

Air University

2017–2019

Bachelor of Science, Electrical Engineering

Air University

2013–2017